EXERCISE-I (Conceptual Questions) — Build Up Your Understanding

ALKANES

1. The order of reactivity of alkyl halides in Wurtz reaction is

(1) R–I > R–Br > R–Cl (2) R–I < R–Br < R–Cl

(3) R–Br > R–I < R–Cl (4) R–I > R–Cl > R–Br

HC0001

2. Kolbe's electrolysis of a mixture of pot. Propanoate and pot. 3-Methylbutanoate gives

(1) Butane and isobutane

(2) Butane and 2,5–dimethylhexane

(3) Butane, 2,5-dimethylhexane and isohexane

(4) Butane and isohexane

HC0002

3. The Corey-House alkane synthesis is carried out by treating an alkyl halide with

(1) Lithium metal

(2) Copper metal

(3) Lithium metal followed by reaction with cuprous iodide and then treating the product with an alkyl halide

(4) Cuprous iodide followed by reaction with alkyl halide

HC0003

4. Which of the following compound is not suitable to obtain from wurtz reaction ?

(1) ethane (2) butane

(3) isobutane (4) hexane

HC0005

5. When ethyl chloride and n-propyl chloride undergoes wurtz reaction which is not obtained

(1) n-butane (2) n-pentane

(3) n-hexane (4) isobutane

HC0006

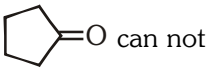
6. $CH_3-\underset{\underset{CH_3}{|}}{CH}\ MgCl + CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-OH \longrightarrow$ 'Q';

What is 'Q' ?

(1) isobutane (2) isopropane

(3) tert. butyl chloride (4) propane

HC0009

7.  can not be converted to  by :

(1) Red P + HI

(2) Wolf Kishner reduction

(3) Clemmensen reaction

(4) $LiAlH_4$

HC0010

8. Which of the following reactions does not involve a C—C bond formation

(1) Hydrolysis of a Grignard reagent

(2) Combination of two alkyl free radicals

(3) Corey-House synthesis of alkanes

(4) $RNa + R-Br \longrightarrow R-R + NaBr$

HC0012

9. Which of the following reactions of methane is incomplete combustion :-

(1) $2CH_4 + O_2 \xrightarrow{Cu/523K/100atm.} 2CH_3OH$

(2) $CH_4 + O_2 \xrightarrow{Mo_2O_3} HCHO + H_2O$

(3) $CH_4 + O_2 \longrightarrow C(s) + 2H_2O(\ell)$

(4) $CH_4 + 2O_2 \longrightarrow CO_2(g) + 2H_2O(\ell)$

HC0016

10. Which is correct about Wurtz reaction ?

(a) It can proceed through free radical mechanism

(b) Alkanes having even no. of C-atom can be prepared

(c) Sodium in Ammonia is used

(d) Sodium in dry ether is used

(1) c, d (2) a, b, d

(3) b,c (4) a, b, c, d

HC0072

11. $C_2H_5-Cl \xrightarrow[H_2]{Zn,\ H^+} (A)$

$(A) \xrightarrow[\text{Dry ether}]{Na} (B) \xleftarrow{H_2/pt} (C) \text{ or } (D)$

The incorrect statement is :

(1) (A) is C_2H_6 (2) (C) can be 1-butene

(3) (A) and (B) are alkane (4) (D) is ethene

HC0073

12. Arrange the following in their boiling points.

(i) n-butane (ii) iso-butane

(iii) n-pentane (iv) iso-pentane

(v) neopentane

(1) iii > i > ii > iv > v (2) v > iv > ii > i > iii

(3) iii > iv > v > i > ii (4) ii > i > v > iv > iii

HC0074

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13. For $CH_3-\overset{O}{\overset{\|}{C}}-ONa \xrightarrow{\text{Electrolysis}}$ (A) alkane

Which is incorrect ?

(1) A is ethane

(2) (A) is formed at anode

(3) CO_2 evolves at cathode

(4) pH near cathode increases during the process

HC0075

ALKENE

14. Ozonolysis of 3–Methyl–1–butene gives a mixture of

(1) Propanal and ethanal

(2) Propanone and ethanal

(3) 2–Methylpropanal and methanal

(4) Butanone and methanal

HC0017

15. Oxidation of isobutylene with acid potassium permanganate gives

(1) Acetone + CO_2 (2) Acetic acid

(3) Acetic acid + CO_2 (4) Acetic acid + acetone

HC0019

16. $CH_3CH_2CH_2Br \xrightarrow[\Delta]{\text{alc.KOH}} A \xrightarrow[(ii)\,Zn,H_2O]{(i)\,O_3} B + C$

In the above reaction A, B and C are given by the set

(1) Propylene, acetone, formaldehyde

(2) Propene, ethanal, methanal

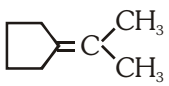
(3) Propyne, acetaldehyde, formaldehyde

(4) Propylene, propionaldehyde, formaldehyde

HC0021

17. Which of the following alkenes on ozonolysis give a mixture of ketones only?

(a) $CH_3-CH=CH-CH_3$ (b) $CH_3-\underset{CH_3}{\underset{|}{CH}}-CH=CH_2$

(c)  (d) $(CH_3)_2C=C\begin{matrix}CH_3\\CH_3\end{matrix}$

(1) a and b (2) b and c

(3) b and d (4) c and d

HC0025

18. Which reaction will not happen at room temperature:

(1) $CH_2=CH_2 \xrightarrow{H_2/Pt} CH_3-CH_3$

(2) $CH_2=CH_2 \xrightarrow{H_2/Pd} CH_3-CH_3$

(3) $CH_2=CH_2 \xrightarrow{H_2/Ni} CH_3-CH_3$

(4) $CH\equiv CH \xrightarrow{H_2/Pt} CH_3-CH_3$

HC0076

19. Which of the following is not electrophilic addition reaction ?

(1) Addition of H^+/H_2O on alkene

(2) Addition of dihydrogen on alkenes

(3) Addition of halogen on alkenes

(4) Addition of hydrogen halides on alkenes

HC0077

20. $CH_3-CH=CH_2 \xrightarrow{HBr} \underset{(A)}{CH_3-\underset{Br}{\underset{|}{CH}}-CH_3} + \underset{(B)}{CH_3-CH_2-CH_2-Br}$

(a) The product A is major

(b) Formation of A follows markovnikov rule

(c) Carbocation formed in A is less stable than that formed in B

(d) Formation of B follows markovnikov rule

The correct statements are :

(1) c, d (2) a, b, c, d

(3) a, b (4) a, d

HC0078

21. For the reaction

$CH_2=CH_2 \xrightarrow[CCl_4]{Br_2} (A) \xrightarrow{\text{alc. KOH}} (B) \xrightarrow{NaNH_2} (C)$

The product (C) is :

(1) $CH_2=CH_2$ (2) $CH_2=\underset{Br}{\underset{|}{CH}}$

(3) H–C≡C–H (4) $\underset{Br}{\underset{|}{CH_2}}-CH_3$

HC0079

22. An alkene A on ozonolysis gives a mixture of ethanal and pentan-3-one. The IUPAC name of A is.

(1) 3-ethyl-3-pentene

(2) 3-ethylidene pentane

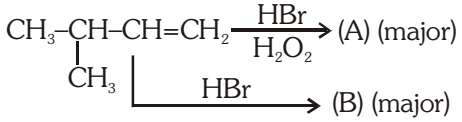
(3) 3-ethyl pent-2-ene

(4) 1,1-diethyl prop-1-ene

HC0080

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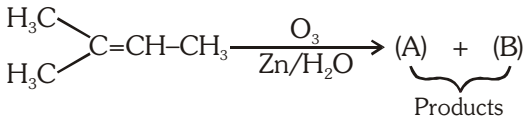
23. For the reaction :



The incorrect statement is :
(1) A and B are chain isomers
(2) A and B are position isomers
(3) A is 1-bromo-3-methyl butane
(4) B is 2-bromo-2-methyl butane

HC0081

24. For the reaction



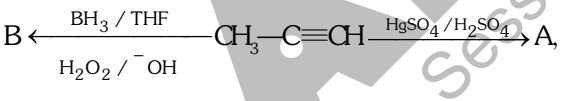
(1) One of the product only show positive tollens test
(2) Both product shows positive tollen's test
(3) Both product shows positive haloform test
(4) Both 1 & 3 are correct

HC0082

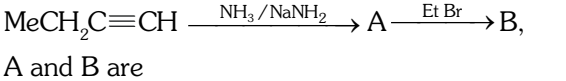
ALKYNES

25. Which of the following compound will not give a precipitate with Tollen's reagent
(1) ethyne (2) 1-butyne
(3) 3-methyl -1-butyne (4) 1-pentene

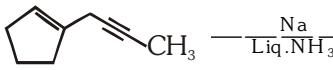
HC0028

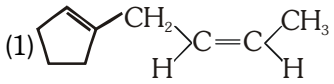
26. 
A and B are
(1) CH_3CH_2CHO, CH_3COCH_3
(2) CH_3COCH_3,CH_3CH_2CHO
(3) CH_3COCH_3 both
(4) CH_3COCH_3, $CH_3CH_2CH_2OH$

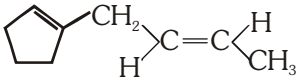
HC0029

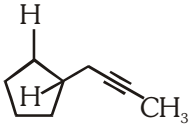
27. 
A and B are
(1) $MeCH_2C{\equiv}CNa$, $MeCH_2C{\equiv}C{-}Et$
(2) $MeCH_2CH{=}CH_2$, $MeCH_2{-}CHEt{-}CH_3$
(3) $MeCH_2CH{=}CHNH_2$,$MeCH_2CH{=}CH{-}NHBr$
(4) $MeCH_2C{\equiv}C{-}NH_2$, $MeC{\equiv}C{-}NH{-}Br$

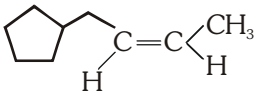
HC0030

28.  product will be :

(1) 

(2) 

(3) 

(4) 

HC0031

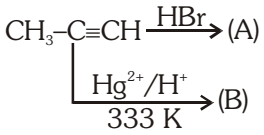
29. To distinguish between propene and propyne, the reagent would be -
(1) Bromine @neetquestionpaper
(2) Alkaline $KMnO_4$
(3) Ammonical silver nitrate
(4) Ozone

HC0032

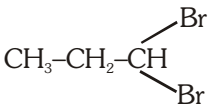
30. The most suitable reagent to differentiate ethyne and ethene is :
(1) Br_2 in CCl_4 (2) $NaHCO_3$
(3) NaOH (4) $NaNH_2$

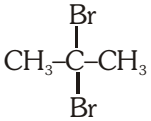
HC0083

31. For the reaction :



Consider statements :-

(a) Product A is 

(b) Product A is 

(c) Product B is $CH_3{-}CH_2{-}CH{=}O$

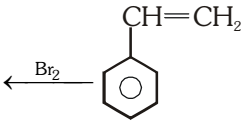
(d) Product B is 

The correct statements are :
(1) b, d (2) a, c
(3) only b (4) only a

HC0084

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AROMATIC HYDROCARBONS

32. A  B ; A and B

respectively are

(1) o-bromo styrene, benzoic acid

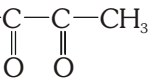
(2) p-bromostyrene, benzaldehyde

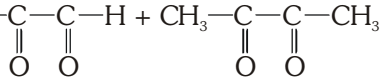
(3) m-bromostyrene, benzaldehyde

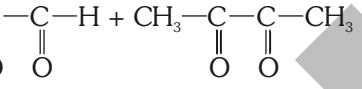
(4) Styrene dibromide, benzoic acid.

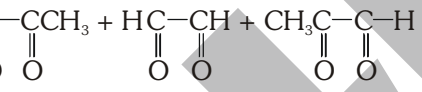
HC0035

33. The ozonolysis product of 1, 2-dimethyl benzene is/are :-

(1) Only 

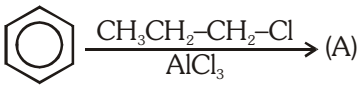
(2) 

(3) 

(4) 

HC0036

34. For the reaction :



Correct statement is :-

(1) A is n-propyl benzene

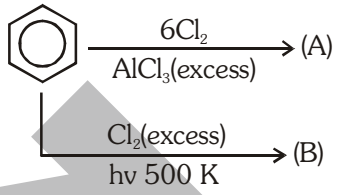
(2) It is a friedel craft alkylation

(3) A is iso propyl benzene

(4) Both 2 & 3

AH0085

35. For the reaction



(1) A is not aromatic

(2) B is aromatic

(3) A is aromatic

(4) B is hexachlorobenzene

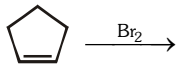
AH0086

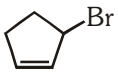
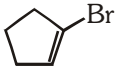
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EXERCISE-I (Conceptual Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**1**	**3**	**3**	**3**	**4**	**4**	**4**	**1**	**3**	**2**	**4**	**3**	**3**	**3**	**1**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**2**	**4**	**3**	**2**	**3**	**3**	**3**	**1**	**4**	**4**	**2**	**1**	**2**	**3**	**4**
Que.	31	32	33	34	35										
Ans.	**1**	**4**	**4**	**4**	**3**										

EXERCISE-III (Analytical Questions) — Master Your Understanding

1.  ; What is the structure of B

(1) 
(2) 
(3) 
(4) 

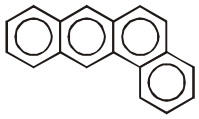
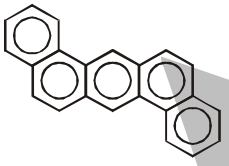
HD0057

2. Which of the following is maximum reactive towards NBS

(1) $PhCH_3$
(2) Ph CH_2—CH_3
(3) $PhCH_2$—CH═CH_2
(4) Ph—CH(—CH_3)—CH═CH_2

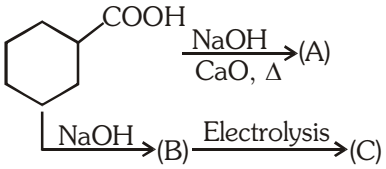
HC0058

3. Which of the following is not carcinogenic ?

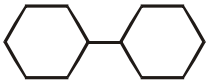
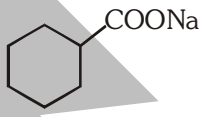
(1) 
(2) 
(3) 
(4) 

HC0092

4. For the reaction given :-



Which is incorrect ?

(1) A is 
(2) C is 
(3) C is 
(4) B is 

HC0093

5. CH_3–C≡C–H $\xrightarrow[1\%\ HgSO_4]{40\%\ H_2SO_4}$ X [Intermediate] $\xrightarrow[\text{Isomerism}]{(y)}$ Z

Identify X, Y and Z

	X	Y	Z
(1)	Prop-1-en-2-ol	Metamerism	Acetone
(2)	Prop-1-en-1-ol	Tautomerism	Propionaldehyde
(3)	Prop-2-en-2-ol	Geometrical isomerism	Acetone
(4)	Prop-1-en-2-ol	Tautomerism	Acetone

HC0094

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EXERCISE-III (Analytical Questions) — ANSWER KEY

Que.	1	2	3	4	5
Ans.	**4**	**4**	**4**	**3**	**4**